

3(a)(i)	any four from: 1 <u>DNA</u> polymerase ; 2 synthesises, complementary / new, DNA strand ; 3 binds to primers on (target) DNA ; 4 adds (DNA) nucleotides to the end of primers or used in extension stage ; 5 (optimum) working temperature of 72 °C ; A 70–75 °C 6 thermostable / does not denature at 95 °C ; 7 (so) can be used for many PCR cycles / does not need replacing ; 8 AVP ; e.g. forming phosphodiester bonds (between adjacent nucleotides)	4
3(a)(ii)	(child) 2 and (child) 3 ;	1
3(a)(iii)	any three from: 1 separates DNA fragments of different, sizes / lengths / mass ; 2 DNA fragments are negatively charged ; 3 (so) DNA fragments move (from cathode) to anode ; 4 smaller DNA fragments migrate, faster / further ; ora 5 DNA fragments from, normal (<i>MT-RNR1</i>) gene travels shorter distance ; ora	4
3(b)	horizontal transmission / transfer plasmids / conjugation / transduction / transfection / transformation ;	1